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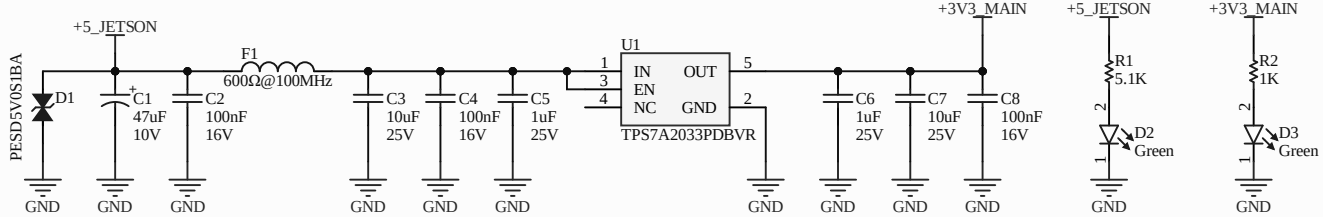

| | |
|---|---|
| Sheet 1 | Title page & Revision |
| Sheet 2 | Power Supply |
| Sheet 3 | Inertial Measurement Unit |
| Sheet 4 | GNSS Receiver |
| Sheet 5 | Controller Area Network |
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# Revision Control

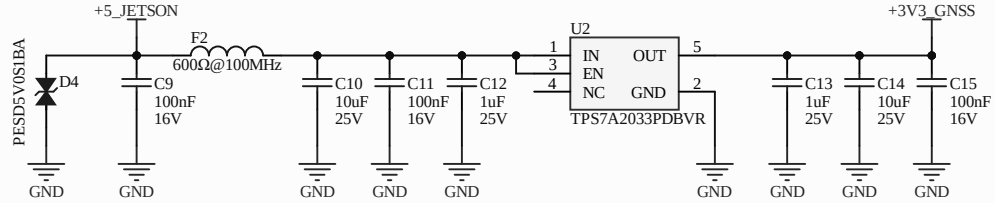
| Version | Description | Date |
|---|---|---|
| 0.0.1 | Initial Release | 10/06/2026 |
| | | |

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|---|---|---|---|
| Title | OrinNav — RTK Navigation HAT for Jetson Orin Nano | | |
| Size: A4 | Number: 1/7 | Revision: 0.0.1 | |
| Date: 10/06/2026 | | Sheet of | |
| File: D:\Electronic\..\Title Block.SchDoc | | Drawn By: Bardia Alikhan Afshar | |

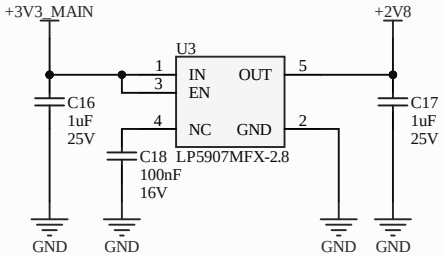
## Main PS - 3.3V/300mA



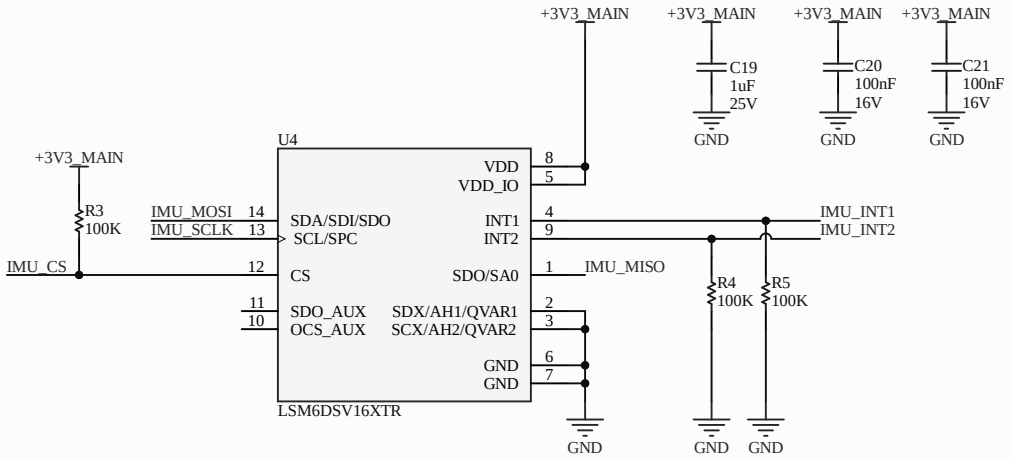

## GNSS PS - 3.3V/300mA



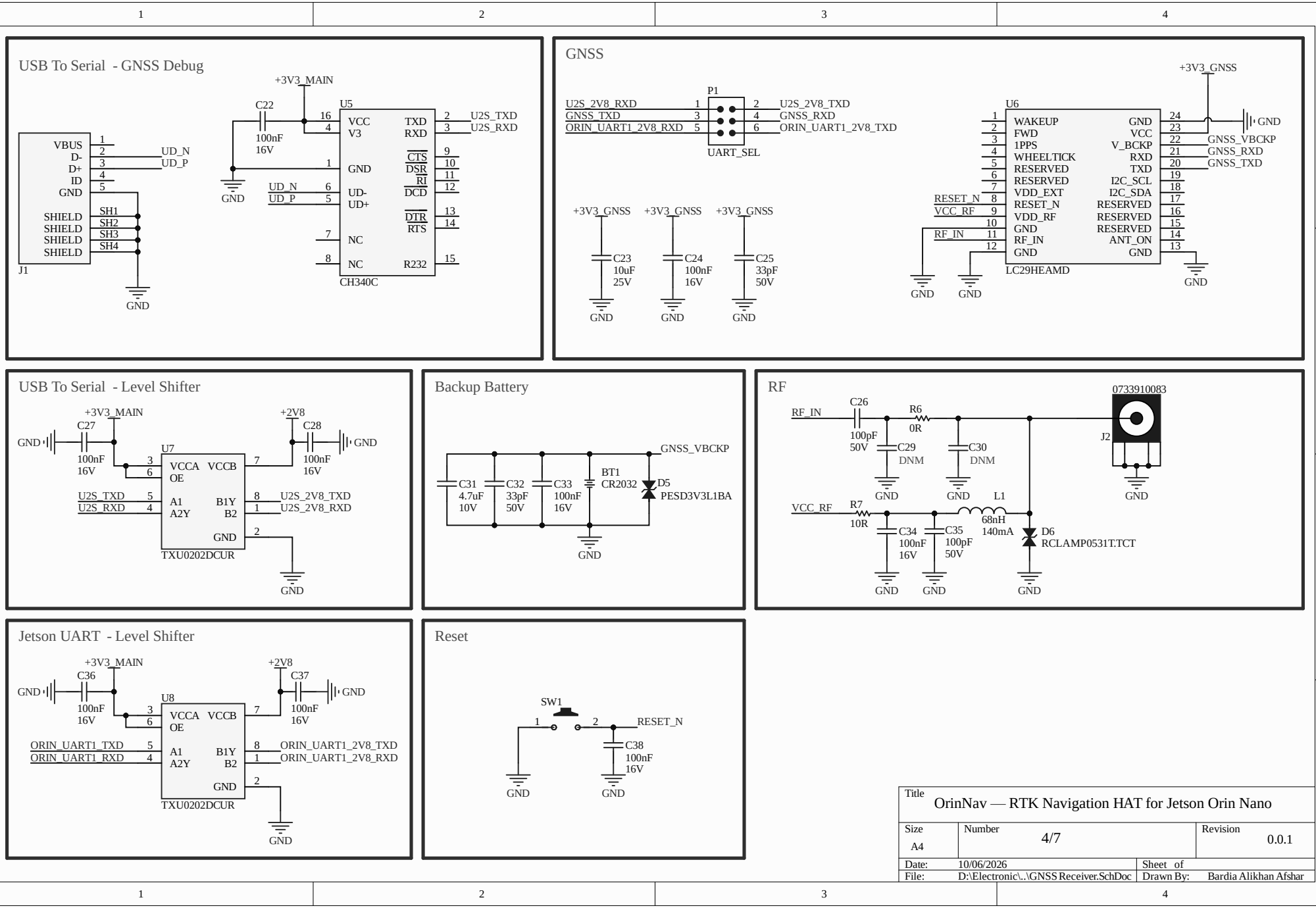

## I/O Shift PS- 2.8V/250mA

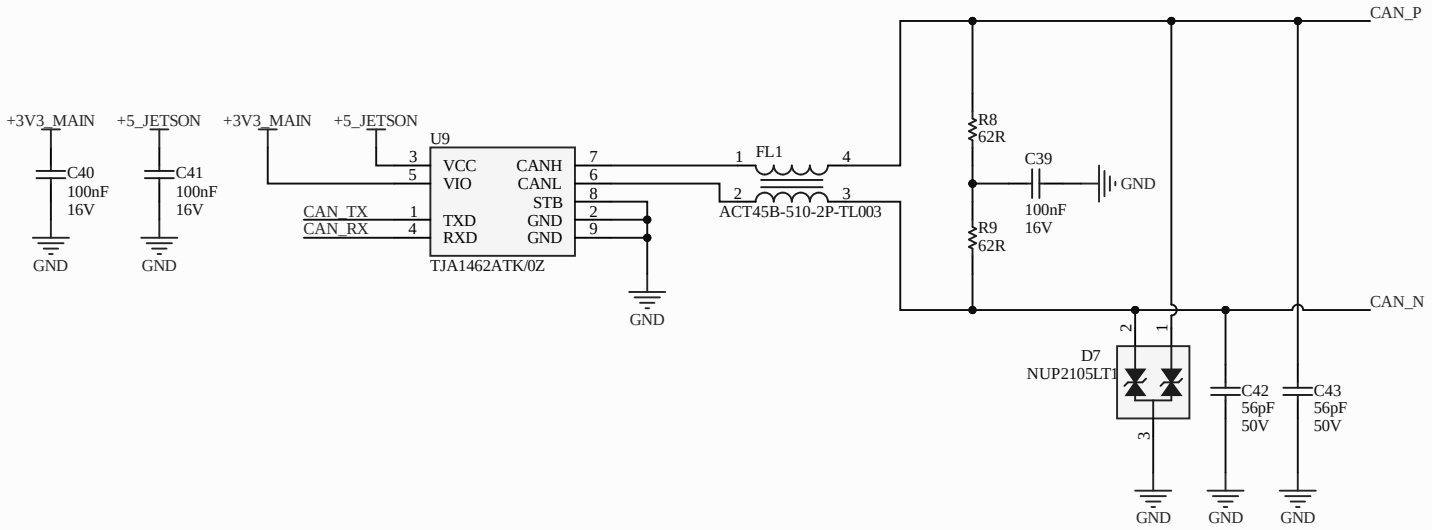



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| Size: A4 | Number: 2/7 | Revision: 0.0.1 |
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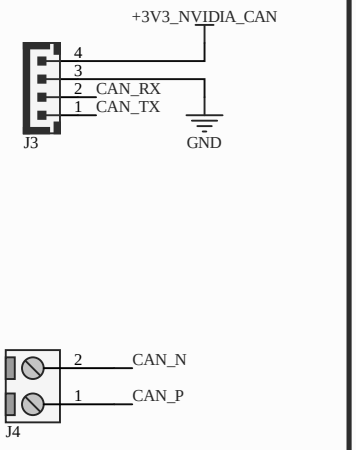

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USB To Serial - GNSS Debug
+3V3_MAIN
C22
100nF
16V
U5
VCC
V3
GND
UD-
UD+
NC
NC
TXD
RXD
CTS
DSR
RI
DCD
DTR
RTS
R232
CH340C
U2S_TXD
U2S_RXD
UD_N
UD_P
VBUS
D-
D+
ID
GND
SHIELD
SHIELD
SHIELD
SHIELD
J1
GND
GNSS
P1
U2S_2V8_RXD
GNSS_TXD
ORIN_UART1_2V8_RXD
U2S_2V8_TXD
GNSS_RXD
ORIN_UART1_2V8_TXD
UART_SEL
+3V3_GNSS
C23
10uF
25V
C24
100nF
16V
C25
33pF
50V
U6
WAKEUP
FWD
1PPS
WHEELTICK
RESERVED
RESERVED
VDD_EXT
RESET_N
VDD_RF
GND
RF_IN
GND
RESET_N
VCC_RF
RF_IN
GND
VCC
V_BCKP
RXD
TXD
I2C_SCL
I2C_SDA
RESERVED
RESERVED
RESERVED
ANT_ON
GND
LC29HEAMD
GNSS_VBCKP
GNSS_RXD
GNSS_TXD
USB To Serial - Level Shifter
+3V3_MAIN
+2V8
C27
100nF
16V
C28
100nF
16V
U7
VCCA
VCCB
OE
A1
A2Y
B1Y
B2
GND
TXU0202DCUR
U2S_TXD
U2S_RXD
U2S_2V8_TXD
U2S_2V8_RXD
Backup Battery
C31
4.7uF
10V
C32
33pF
50V
C33
100nF
16V
BT1
CR2032
D5
PESD3V3L1BA
GNSS_VBCKP
RF
RF_IN
C26
100pF
50V
R6
0R
C29
DNM
C30
DNM
J2
0733910083
VCC_RF
R7
10R
C34
100nF
16V
C35
100pF
50V
L1
68nH
140mA
D6
RCLAMP0531T.TCT
Jetson UART - Level Shifter
+3V3_MAIN
+2V8
C36
100nF
16V
C37
100nF
16V
U8
VCCA
VCCB
OE
A1
A2Y
B1Y
B2
GND
TXU0202DCUR
ORIN_UART1_TXD
ORIN_UART1_RXD
ORIN_UART1_2V8_TXD
ORIN_UART1_2V8_RXD
Reset
SW1
RESET_N
C38
100nF
16V
Title
OrinNav — RTK Navigation HAT for Jetson Orin Nano
Size
A4
Number
4/7
Revision
0.0.1
Date:
10/06/2026
Sheet of
File:
D:\Electronic\..\GNSS Receiver.SchDoc
Drawn By:
Bardia Alikhan Afshar

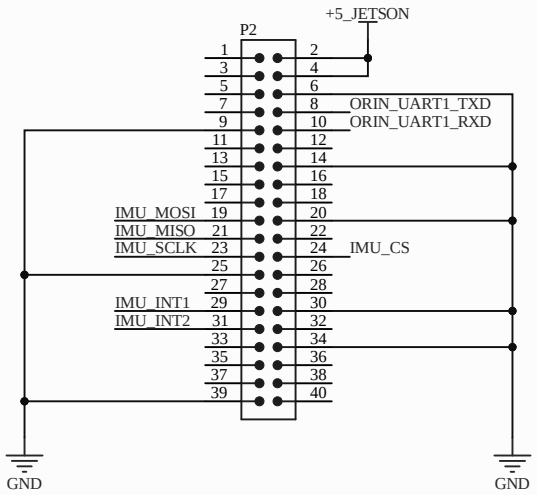



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| Title | OrinNav — RTK Navigation HAT for Jetson Orin Nano | | |
| Size: A4 | Number: 5/7 | Revision: 0.0.1 | |
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| File: D:\Electronic\..\CAN Interface.SchDoc | | Drawn By: Bardia Alikhan Afshar | |

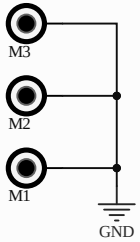
## CAN




## Jetson Orin Nano 40-pin Header




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|---|---|---|
| Title: OrinNav — RTK Navigation HAT for Jetson Orin Nano | | |
| Size: A4 | Number: 6/7 | Revision: 0.0.1 |
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|---|---|---|---|
| Title | OrinNav — RTK Navigation HAT for Jetson Orin Nano | | |
| Size: A4 | Number: 7/7 | | Revision: 0.0.1 |
| Date: 10/06/2026 | | Sheet of | |
| File: D:\Electronic\..\Mechanical.SchDoc | | Drawn By: | Bardia Alikhan Afshar |